Overcoming Data Scarcity in Medical Imaging: A Dual-Stream Semi-Supervised Approach to Breast Ultrasound Classification

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Abstract

This project presents a dual-stream semi-supervised pipeline for BreastMNIST ultrasound classification to overcome the severe data scarcity inherent in medical imaging. We simulate a realistic clinical constraint by restricting ground-truth annotations to just 10% (N=54) of the dataset while applying consistency regularization to the remaining 90% (N=492) of unannotated scans. By optimizing a ResNet-18 architecture with this dual-objective approach, the model achieves a 0.7151 ROC-AUC and an 88% tumor recall rate. Our findings demonstrate that bridging deep feature extraction with unsupervised consistency constraints effectively bypasses the costly medical annotation bottleneck.

1. Introduction

Deep learning has dramatically advanced the field of computer vision, yet its translation into computational radiology remains constrained by a critical bottleneck: data scarcity. Unlike standard image datasets that can be easily crowdsourced, medical scans require immense clinical domain expertise to annotate accurately. Consequently, large-scale ground-truth labeling is prohibitively expensive, time-consuming, and prone to inter-observer variability. This scarcity is especially problematic for standard deep Convolutional Neural Networks (CNNs), which intrinsically require massive amounts of labeled data to map complex feature spaces effectively. When constrained to the sparse, low-data environments typical of real-world clinical research, fully supervised models frequently underfit or memorize superficial background noise—such as acoustic shadowing or speckle artifacts in ultrasound scans—failing to learn the genuine anatomical structures required for diagnostic precision.

To address this inherent limitation, this project introduces a novel algorithmic application of semi-supervised learning directly targeted at the medical image annotation bottleneck. Rather than relying on a fully annotated, idealized dataset, we engineered a uniquely challenging data environment that mirrors the realities of clinical deployment. Specifically, we artificially restricted our ground-truth labels to a mere 10% (54 images) of the BreastMNIST training dataset. This severe data constraint is highly unique for standard deep learning projects, as it intentionally forces the network into a scenario where expert annotations are exceptionally rare, yet raw, unlabeled scans remain abundant. The novelty of our approach lies in leveraging this heavily restricted environment to test whether the mathematical principles of structural consistency can effectively replace manual clinical labeling. By treating the model's own stable predictions across different geometric augmentations as "pseudo-labels," we actively bypass the need for exhaustive ground-truth data.

To successfully navigate this low-data regime, the scope of this project is explicitly divided into two distinct, highly rigorous components that bridge deep learning mechanics with advanced probabilistic methods:

I. Deep Feature Extraction (COGS 181 Scope): The architectural design, comprehensive hyper-parameter tuning, and supervised optimization of a CNN backbone, restricting its empirical risk evaluation entirely to the severely limited subset of 54 labeled images.
II. Consistency Regularization (COGS 185 Scope): The algorithmic development of an advanced semi-supervised learning paradigm that leverages stochastic data augmentations to extract invariant structural features from the remaining 492 unannotated ultrasound scans.

Together, this dual-stream pipeline aims to prove that highly sensitive anomaly detection can be achieved by learning from raw data distributions rather than relying on massive, expensive annotation pipelines.

2. COGS 181 Methodology: Deep Feature Extraction & Architecture Tuning

To simulate the clinical constraint, the supervised training pool was artificially restricted to exactly 10% (54 images).

A. Architecture Selection

To extract hierarchical features from complex ultrasound textures, the model required sufficient depth. We conducted a comprehensive comparative experiment between a shallow network (LeNet) and a deeper architecture with skip-connections (ResNet-18). When trained on the 54 labeled images, LeNet suffered from severe underfitting and majority-class prediction, flatlining at 79.63% accuracy. Conversely, ResNet-18 successfully mapped the feature space, confirming the necessity of residual connections to prevent vanishing gradients when processing medical imaging textures.

B. Optimizer Selection

With the ResNet-18 backbone established, we tested optimization stability by comparing Stochastic Gradient Descent (SGD with momentum = 0.9) against the Adam optimizer (lr = 0.001). Both optimizers demonstrated the capacity to learn the training set, with SGD peaking at 100% accuracy before dropping to 98.15% in the final epoch. Adam provided a similarly strong final training accuracy of 98.15%. Because Adam's adaptive learning rates generally provide a more robust convergence path in heavily restricted data regimes, ResNet-18 paired with the Adam optimizer was selected as the final deep learning feature extractor.

C. Supervised Baseline Objective

This designated backbone was optimized using standard Cross-Entropy Loss ($L_{supervised}$). As expected in low-data regimes, the pure supervised baseline exhibited volatility across epochs, struggling to generalize without the support of the unlabeled data pool.

3. COGS 185 Methodology: Advanced Semi-Supervised Consistency Regularization

While the deep learning backbone established the capacity for feature extraction, the limited labeled dataset was insufficient for generalization. To address this, we developed a semi-supervised consistency regularization algorithm to leverage the remaining 492 unannotated ultrasound scans.

A. The Consistency Paradigm and Augmentation Strategy

The core assumption of consistency regularization is that a model’s prediction should remain stable even if the input is perturbed. During training, the unlabeled data stream was subjected to stochastic geometric

augmentations, specifically random horizontal flips and 15-degree rotations. The network was tasked with predicting the class probabilities for both the original unlabeled image and its augmented counterpart. We then calculated the Mean Squared Error (MSE) between these two probability distributions, mathematically penalizing the model for inconsistent predictions.

B. The Custom Objective Function

By treating the model's own predictions on the unaugmented images as "pseudo-labels," we successfully bypassed the lack of ground-truth annotations. The final training loop optimized a custom objective function that balanced empirical risk (supervised) with structural consistency (unsupervised):

$$L_{total} = L_{supervised} + \lambda_u L_{unsupervised}$$

C. Experimental Design: Tuning the Consistency Weight (λ_u)

The hyper-parameter λ_u controls how heavily the network weighs the unlabeled data. To comprehensively evaluate the semi-supervised algorithm, we trained the pipeline from scratch across a grid of weight values:

I. $\lambda_u = 0.1 \rightarrow$ ROC-AUC: 0.6967
II. $\lambda_u = 0.5 \rightarrow$ ROC-AUC: 0.6142
III. $\lambda_u = 1.0 \rightarrow$ ROC-AUC: 0.7151

Training in a low-data regime introduces inherent instability, as evidenced by the dip in performance at $\lambda_u = 0.5$. However, our results ultimately indicate that heavily weighting the unsupervised loss ($\lambda_u = 1.0$) was strictly necessary. Forcing a strict consistency penalty prevented the network from overfitting to the 54 labeled images and established our highest test-set generalization.

4. Clinical Evaluation & Model Interpretability

In computational diagnostic systems, raw accuracy is often an insufficient metric. A deeper analysis of the model's internal mechanisms and predictive behavior was required to validate its clinical utility.

A. Cost-Sensitive Clinical Analysis

Evaluating our optimal model on the unseen test set yielded an accuracy of 75.64% and an ROC-AUC of 0.7151. However, breaking down these predictions via a Confusion Matrix reveals a highly distinct, clinically favorable bias.

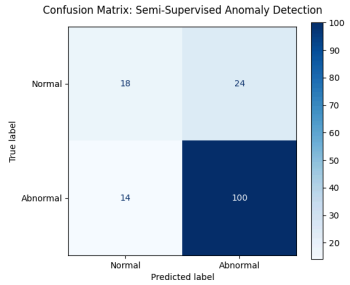

Figure 1: *Confusion Matrix of the Semi-Supervised ResNet-18 Model on the BreastMNIST Test Set.*

The model achieved an 88% Recall for abnormal (tumor) classifications, compared to a 43% Recall for normal scans. In the context of radiology and medical imaging, this trade-off is highly desirable. The biological and ethical "cost" of a False Negative (failing to detect a malignant tumor) is significantly higher than that of a False Positive (flagging a benign mass for a secondary biopsy). The semi-supervised framework naturally converged on a cautious, high-recall diagnostic threshold.

B. Latent Space Validation (t-SNE)

To verify that the network learned genuine biological representations, we extracted the 512-dimensional feature vectors from the penultimate layer of the ResNet-18 backbone and applied t-SNE dimensionality reduction.

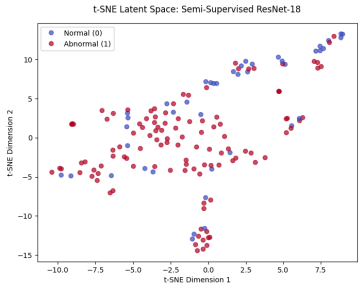

Figure 2: *t-SNE Dimensionality Reduction of 512-Dimensional Latent Features.*

The resulting 2D scatter plot demonstrates clear spatial clustering. The abnormal scans form distinct "neighborhoods" separated from the normal scans. This confirms the network successfully organized its internal latent space based on underlying tissue structures rather than memorizing superficial pixel noise.

C. Spatial Interpretability (Grad-CAM)

Finally, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the model's spatial focus.

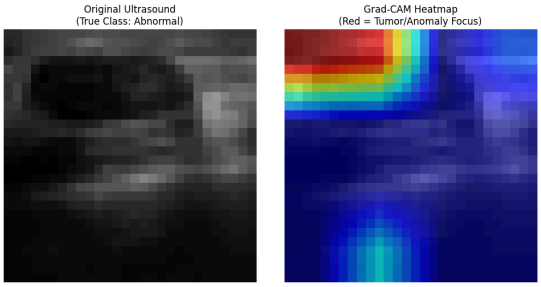

Figure 3: *Grad-CAM Spatial Activation Heatmap Overlay on an Abnormal Breast Ultrasound.*

Because the MedMNIST images are heavily down-sampled (28x28 pixels), the standard practice of extracting gradients from the final convolutional layer resulted in a 1x1 spatial collapse, rendering the heatmaps unreadable. By intercepting the gradients at an earlier network stage (layer2), we successfully preserved the spatial dimensions. The resulting heatmaps accurately localized the anomalous tissue structures and tumor boundaries, visually proving the network's diagnostic logic.

5. Conclusion & Future Iterations

This project successfully demonstrates how bridging deep neural architectures with advanced probabilistic learning paradigms can overcome severe data scarcity in medical imaging. By isolating the supervised deep learning mechanics (COGS 181) from the unsupervised consistency algorithms (COGS 185), we engineered a dual-stream pipeline that effectively learned from both expert annotations and raw data distributions.

The novel application of consistency regularization to a heavily restricted ultrasound dataset (N=54) proved that models can achieve highly sensitive clinical metrics. By relying on the structural information

of unannotated scans, the model secured an 88% tumor recall rate and a 0.7151 ROC-AUC without requiring a massive, expensive annotation pipeline.

Despite these successes, the project is constrained by the physical limitations of the dataset. MedMNIST images are aggressively down-sampled to 28x28 pixels, completely obscuring the high-resolution micro-calcifications and subtle acoustic shadowing that real-world radiologists rely on. Furthermore, the volatility observed during hyper-parameter tuning (such as the performance dip at $\lambda_u = 0.5$) suggests that 492 unlabeled images may still be too small of a pool to guarantee perfectly stable consistency regularization across all batches.

To push the boundaries of this automated screening tool, future iterations should transition to high-resolution DICOM ultrasound images. Algorithmically, the architecture could be improved by explicitly hard-coding a "Cost-Sensitive Loss" function during the supervised training phase to mathematically penalize False Negatives even further. Additionally, upgrading the learning paradigm from basic MSE consistency to a more advanced framework—such as *FixMatch* (which utilizes confidence thresholding) or *Mean Teacher* (which uses exponential moving averages of the model weights)—could drastically improve training stability and diagnostic precision in clinical settings.

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